The Capital Velocity Gap:

A Multi-State Forensic Analysis of Innovation Funding Intermediaries in the Rust Belt

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Abstract

State governments allocate substantial public revenue to foster entrepreneurship, typically via nonprofit 'innovation intermediaries' and state-backed venture capital programs. This paper introduces the Direct Capital Ratio (DCR)—the proportion of funding reaching founders as investment—as a metric for institutional efficiency. A forensic audit of intermediaries across the Rust Belt (PA, OH, MI, WI, IN) reveals a structural efficiency crisis: DCRs range from under 1% to 55%, with an average of approximately 27 cents reaching founders for every state dollar appropriated.

Pennsylvania presents a particularly instructive case of architectural fragmentation: state innovation capital flows through two parallel tracks—four regional Ben Franklin Technology Partners centers and approximately two dozen locally-based venture capital firms—with no coordination, no shared outcome accountability, and no customer procurement requirements. This dual-track system, combined, underperforms Indiana's single lean deployment vehicle (Elevate Ventures) on both capital efficiency and customer access metrics.

Concurrently, recent field evidence from Kim, Kim, and Koning (2026) indicates that AI-enabled startups grow 1.9x faster while requiring 39.5% less external capital, fundamentally challenging the value proposition of high-overhead mentorship models. I propose a three-part reform: a 75% Direct Capital Mandate, a First Customer Clause requiring pilot commitments from tax-incentivized corporations, and consolidation of fragmented funding architectures.

1. Introduction

The 'intermediary model' of innovation funding—bundling capital with mentorship and technical assistance—originated in the low-information environment of the 1980s. Pennsylvania's Ben Franklin Technology Partnership, established in 1983, became the national template for this approach, predicated on the assumption that early-stage founders lack both capital and the knowledge to deploy it effectively. Over time, states have layered additional mechanisms onto this foundation, including state-backed venture capital programs, creating parallel funding tracks with independent overhead structures and no unified outcome accountability.

Current forensic data suggests these organizations have evolved into self-perpetuating overhead machines. This paper utilizes the Direct Capital Ratio to evaluate the efficiency of the intermediary model across five Rust Belt states, identifying two systemic failures: first, an 'overhead trap' where administrative costs consistently decouple from founder capital deployment; and second, an 'architectural fragmentation' problem where capital is dispersed across multiple entities with duplicative costs and no coordination.

The analysis reveals that on average, only 27 cents of every state innovation dollar reaches founders as direct capital. In Pennsylvania, where capital flows through both nonprofit intermediaries and state-backed VCs, the combined apparatus underperforms Indiana's single consolidated vehicle on every efficiency metric examined.

2. Methodology

2.1 The Direct Capital Ratio

The Direct Capital Ratio (DCR) measures the efficiency of capital transmission from the state to the entrepreneur. To ensure academic rigor, this metric relies strictly on audited IRS Form 990 Part IX (Statement of Functional Expenses) and Federal Audit Clearinghouse Single Audit data.

The formula is defined as:

$$DCR = \text{Direct Capital Disbursed to Founders} / \text{Total State Funding Received}$$

Numerator: Includes direct grants, equity investments, and convertible notes disbursed to founding teams.

Denominator: Total state-designated appropriations received by the intermediary, excluding federal passthrough funds.

Exclusions: Salaries, 'technical assistance,' consulting fees, rent, and general administrative expenses are categorized strictly as overhead, regardless of how intermediaries characterize them in program reporting. For venture capital recipients, standard management fees (typically 2% annually) and operational costs are similarly categorized as overhead.

2.2 AI-Assisted Research Methodology

This analysis was conducted using AI-assisted research methods, employing both Anthropic's Claude and Google's Gemini for data extraction, cross-validation, and analysis structuring. The complete research cycle—from initial 990 extraction through multi-state comparative analysis to policy recommendation—was completed in approximately one hour by a single researcher.

This methodological approach is itself relevant to the paper's thesis. The same AI capabilities that enable a single researcher to conduct forensic audits previously requiring institutional research teams are the capabilities that reduce founders' dependence on intermediary guidance services. The research methodology demonstrates the obsolescence argument it documents.

All AI-generated outputs were cross-validated across models and verified against primary source documents (IRS Form 990 filings accessed via ProPublica Nonprofit Explorer; Federal Audit Clearinghouse SEFA data; state program documentation). The author takes full responsibility for the accuracy of all findings and conclusions.

3. Findings: Multi-State Forensic Benchmarks (FY2024–2025)

3.1 Pennsylvania: The Fragmentation Problem

Pennsylvania presents the most instructive case of architectural dysfunction. State innovation capital flows through two parallel tracks operating with independent overhead structures, no coordination mechanism, and no unified outcome accountability:

Track 1 — Ben Franklin Technology Partners: Four regional nonprofit centers (Innovation Works in Pittsburgh, BFTP-SEP in Philadelphia, BFTP-NEP in Bethlehem, and BFTP-CNP in State College) receive state appropriations to operate accelerators, provide 'technical assistance,' and make seed investments.

Track 2 — State-Backed Venture Capital: Approximately two dozen locally-based VC firms receive state capital allocations, each charging standard management fees (typically 2% of AUM annually) plus carried interest, with independent reporting and no coordination with the BFTP track.

This dual-track architecture creates compounding overhead: founders seeking state-backed capital must navigate multiple entities, each extracting administrative costs, with no single point of accountability for outcomes. The state pays intermediary salaries at BFTP centers AND management fees to VC firms, while neither track is required to deliver customers to the founders they fund.

3.1.1 Ben Franklin Technology Partners Network

The BFTP network demonstrates that inefficiency is a structural feature of regional centers, though the degree of 'overhead absorption' varies significantly.

Center	Location	DCR	Primary Finding
Innovation Works	Pittsburgh	25%	Staff compensation ($7M+) now exceeds total disbursements to founders.
BFTP-SEP	Philadelphia	0.2%	Management overhead almost entirely consumes state funding before it reaches founders.
BFTP-NEP	Bethlehem	~36%	Approved $5.28M for 86 companies in FY2025, yet maintains heavy incubator overhead.
BFTP-CNP	State College	~30%	Direct seed deployment constrained by high mentorship-to-capital ratios.

Table 1: Direct Capital Ratios across Pennsylvania's Ben Franklin Technology Partners Network (FY2025)

The observed 0.2% DCR at BFTP-SEP represents a near-total absorption of public capital by organizational overhead, providing the strongest evidence for what this paper terms the 'dead

money' thesis—the systematic retention of innovation funding within intermediary cost structures rather than deployment to founders.

3.1.2 The Insurance Premium Tax Credit Mechanism

Pennsylvania's Innovate in PA program (Act 52 of 2013) provides additional context for the capital efficiency problem. The state sold $100 million in deferred insurance premium tax credits to 29 qualified insurance companies at 87 cents on the dollar to generate upfront cash for intermediaries. Critically, the statute contains no requirement that participating insurers provide pilot contracts or 'first customer' status to funded startups. The credits offset insurer tax liability while cash flows through the intermediary layer to fund accelerator operations—a pure 'capital supply' mechanism with no 'customer demand' component.

Alpha Lab Health, Pittsburgh's primary attempt to bridge the capital-to-customer gap, operates as a collaboration between Innovation Works and Allegheny Health Network (AHN), the provider arm of Highmark. While AHN provides 'clinical readiness' and mentorship, public data on pilot-to-contract conversion rates is not disclosed. The three-year Pennsylvania headquarters retention rate for program graduates similarly remains an internal metric unavailable to policy researchers.

3.2 Ohio: The Transparency Gap

Ohio's intermediaries show a consistent '20% Floor,' where approximately 80 cents of every dollar is retained for internal operations.

JumpStart Inc. (Cleveland): Documented DCR of 21%. The organization reports high 'consultant expenses' which often represent internal service credits rather than liquid capital disbursed to founders.

Rev1 Ventures (Columbus): FY2024 Form 990 extractions show $11.1M in total expenses against a deployment ratio estimated at approximately 24%, with executive compensation ($785K) representing a significant secondary overhead metric.

Ohio utilizes its substantial insurance sector presence (Nationwide, Progressive) to fund development through Opportunity Zone Tax Credits applicable against insurance premium taxes (ORC 5725.18). However, as in Pennsylvania, there is no linkage between tax incentives and

corporate procurement from funded startups. While Nationwide and KeyCorp maintain formal 'Innovation' arms, these operate as traditional venture/accelerator models rather than 'first customer' programs mandated by statute.

3.3 Michigan: High-Cost Intermediation

Invest Detroit: FY2024 total expenses reached $22.2M, with $7.1M in salaries representing 32.3% of expenses. Deployment remains higher than Pennsylvania peers, with a DCR of approximately 45%.

Ann Arbor SPARK: FY2024 Form 990 extractions show $11.3M in total expenses. While SPARK assisted 457 startups, direct investment from managed funds totaled $2.3M, yielding a calculated DCR of 20.3%.

Michigan's recently established $60 million Innovation Fund leans on automotive anchors rather than the insurance-heavy model of Pennsylvania and Ohio. Ford's Michigan Central Station restoration serves as a physical pilot site for mobility startups. However, there is no evidence of a statutory 'buy local' requirement for companies receiving R&D tax credits; procurement remains a private corporate decision.

3.4 Indiana: The Consolidated Benchmark

Indiana presents the regional proof point that consolidated, high-velocity deployment is operationally feasible.

Elevate Ventures: Operates as a single statewide vehicle with a model that decouples annual expenses from capital velocity. In 2024, Elevate maintained $10.8M in expenses while committing $18.9M in capital, achieving a DCR of 55%—more than double the Pennsylvania BFTP network average.

Customer Access Infrastructure: Through the Central Indiana Corporate Partnership (CICP) and AnalytiXIN, corporate anchors including Eli Lilly, Anthem, and Cummins have created formal academic-industry data hubs. Unlike Pennsylvania's insurance premium tax credit model, Indiana's approach builds explicit pilot pipelines between funded startups and corporate customers.

Anthem's 'studio' specifically develops products for deployment within their own ecosystem, creating a direct path from funding to revenue.

Tax Credit Alignment: Indiana's $20M Venture Capital Investment (VCI) Tax Credit is transferable and can be claimed against insurance premium tax, creating a financial link between insurers and startups. However, unlike Pennsylvania's structure, this operates alongside—not instead of—direct corporate pilot commitments.

Dimension	Pennsylvania (Dual-Track)	Indiana (Consolidated)
Funding Architecture	4 BFTP centers + ~24 VC firms; no coordination	1 statewide vehicle (Elevate Ventures)
Direct Capital Ratio	0.2% to 36% (BFTP avg ~23%)	55%
Customer Access	No procurement requirement; pilot data not public	CICP/AnalytiXIN formal pilot programs with Lilly, Anthem, Cummins
Outcome Accountability	Fragmented reporting; retention data not disclosed	Unified reporting through single entity
Overhead Structure	Duplicative: BFTP salaries + VC management fees	Single overhead layer

Table 2: Architectural Comparison: Pennsylvania vs. Indiana Innovation Funding Systems

3.5 Wisconsin: Replicating the Intermediary Problem

WEDC: Analysis of the Wisconsin Economic Development Corporation remains preliminary due to its status as a quasi-public body with different reporting requirements. FY2025 reports indicate $197.7M in total contracts, yet only $39.1M were direct grants to businesses, suggesting an initial DCR of approximately 20%.

Wisconsin's newest legislation creates a Community Development Entity (CDE) Investment Credit specifically for insurance companies. While this encourages capital flow from insurers into rural and urban low-income community businesses, it lacks any 'first customer' or pilot contract requirement—replicating the intermediary problem by adding a CDE layer between insurance tax dollars and operating businesses.

4. The AI Inflection: The Obsolescence of 'Guidance'

The traditional justification for intermediary overhead is that founders need 'technical assistance' and mentorship to survive the discovery phase of company building. However, recent field evidence challenges this assumption fundamentally.

Kim, Kim, and Koning (2026) conducted a randomized field experiment with 515 high-growth startups, finding that those utilizing AI for strategy, market research, and operational discovery grew 1.9x faster while reducing their demand for external capital by 39.5%. The implications for intermediary models are stark: if AI allows founders to navigate the 'discovery problem' at near-zero marginal cost, the high-cost mentorship and guidance functions of intermediaries are no longer economically justifiable.

The methodology of this paper provides additional evidence for this thesis. A single researcher using AI tools conducted a multi-state forensic analysis in one hour—work that would traditionally require a research team, months of data extraction, and significant institutional resources. If AI enables this level of analytical capability for policy research, the same capabilities are available to founders navigating market discovery, competitive analysis, and strategic planning. The technology intermediaries are tasked with helping founders adopt is the same technology that makes the intermediary model obsolete.

5. Policy Recommendations

Based on the foregoing analysis, I propose a three-part structural reform for state innovation funding:

5.1 The 75% Direct Capital Mandate

Require that at least 75% of state innovation funding be disbursed directly to founders as capital.

Rationale: A 75% DCR floor would nearly triple the capital available to founders without requiring any increase in state appropriations. For every $10 million allocated, $7.5 million would reach founders as investment capital, compared to approximately $2.7 million under current structures.

Implementation: State contracts with intermediaries should include a 'Direct Capital Floor' provision, with compliance verifiable via annual Federal Audit Clearinghouse filings.

5.2 The First Customer Clause

Require corporations receiving state tax incentives above a threshold (e.g., $10 million annually) to pilot at least one product per year from a state-headquartered startup.

Rationale: Capital without customers funds founder relocation, not ecosystem development. The Indiana model demonstrates that corporate pilot commitments—not just funding—are necessary to retain founders in-state. Pennsylvania's insurance premium tax credit mechanism delivers tax benefits to major insurers with no reciprocal procurement obligation.

Implementation: Attach procurement requirements to existing tax incentive programs. Compliance could be verified through annual reporting to the state economic development agency, with pilot contract documentation required.

5.3 Architectural Consolidation

Consolidate fragmented funding tracks into a single high-velocity deployment vehicle, or at minimum require unified outcome reporting across all state-backed capital sources.

Rationale: Pennsylvania's dual-track system—BFTP centers plus state-backed VCs—creates duplicative overhead with no coordination. The state pays intermediary salaries AND VC management fees while neither track is accountable for unified outcomes. Indiana's single-vehicle model (Elevate Ventures) achieves 55% DCR with unified reporting.

Implementation: Either merge existing programs into a single statewide deployment entity, or mandate common outcome metrics (DCR, pilot-to-contract conversion, three-year founder retention) across all state-backed capital programs with public reporting.

6. Conclusion

The architecture of state innovation funding in the Rust Belt has decoupled from its original mission. Organizations and programs designed to accelerate founder success have evolved into overhead-absorbing entities that retain the majority of public capital while providing no pathway to first customers.

Pennsylvania's dual-track system—fragmenting capital across nonprofit intermediaries and state-backed venture capital firms—exemplifies the structural dysfunction. Combined, these apparatuses underperform Indiana's single consolidated vehicle on every efficiency metric examined. The problem is not the quantum of funding; it is the architecture through which funding flows.

By adopting a 75% Direct Capital Mandate, a First Customer Clause, and architectural consolidation, states can ensure that public innovation funding compounds in the hands of founders rather than administrators—accelerating job creation, retaining talent in-region, and generating higher returns on public investment. The policy ask is simple: let founders build, and give them customers to build for.

Acknowledgments

This research was conducted with AI assistance from Anthropic's Claude and Google's Gemini for data extraction, analysis structuring, cross-validation, and document preparation. All outputs were verified against primary source documents.

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